

30% Evaluation Checklist for AI-Based Web/ Mobile Application

This 30% Evaluation Checklist may be used for 3 different areas of project based on AI.

Model-Centric Approach

Focus: Major emphasis on developing and improving the AI model, with minimal emphasis on application development.

1. Proposal Refinement and Validation (5 Marks)

- Clear and detailed objectives for the AI model development.
- Explanation of how the AI model will be integrated into a minimal application.
- Feasibility analysis for the model's architecture and expected outcomes.
- Updated proposal incorporating feedback from the defense.

2. AI Model Development (10 Marks)

- **Core AI Model Development:**
 - Initial implementation with basic functionality (e.g., image classification, sentiment analysis).
- **Preliminary Training Results:**
 - Model trained on sample data with metrics like accuracy or precision.
 - Explanation of chosen algorithm/model (e.g., CNN, LSTM).
- **Data Preprocessing Pipeline:**
 - Initial implementation of data cleaning workflows (e.g., handling missing values, normalization).
- **Integration Readiness:**
 - AI module prepared for integration (e.g., as REST API or on-device model).

3. Application Development (5 Marks)

- **Minimal Functional UI:**
 - Basic interface for testing the AI model (e.g., input form, prediction display).
- **Basic Backend:**
 - Simple backend setup for model input-output handling.

Proposed Modules

- **AI Model Integration Module:** Handles core AI functionality (e.g., model inputs & outputs).

4. Integration with Information System (5 Marks)

- **Initial Data Flow Implementation:**
 - Partial implementation of data flow (mobile app → AI module → backend).
- **Mock Information System Setup:**
 - Functional mock or partial backend system (e.g., database for AI outputs).
- **Workflow Design for Integration:**
 - High-level workflow showing interactions between mobile app, AI module, and backend.

5. Documentation and Technical Design (5 Marks)

- **Updated System Architecture Diagram:**
 - Diagram showing integration of mobile app, AI module, and information system.
- **Workflow and Sequence Diagrams:**
 - Step-by-step visualization of data flow and interactions.
- **Updated Requirements Document:**
 - Refined functional and non-functional requirements post-proposal defense.

6. Prototype Demonstration (5 Marks)

- **Partially Functional Prototype:**
 - Demonstrates basic functionality of the mobile app, AI module, and backend.
 - Example: Upload image → AI predicts → Prediction saved to backend.
- **Working Data Flow:**
 - End-to-end demo of initial data flow between system components.

7. Project Management (5 Marks)

Timeline Adherence:

- Milestones achieved as planned for the first 30% phase.
- Evidence of timely completion of key deliverables.

Collaboration and Communication:

- Regular meetings and feedback with supervisors documented.
- Example: Meeting notes, supervisor approvals, or comments on progress.

Use of Project Tracking Tools:

- Use of tools like Jira, Trello, or GitHub for issue tracking and progress updates.

Application-Centric Approach

Focus: Major emphasis on developing the application with minimal focus on the AI model.

1. Proposal Refinement and Validation (5 Marks)

- Clear objectives for application design and integration of AI as a feature.
- Feasibility analysis for the application workflow and user interaction.
- Updated proposal incorporating feedback from the defense.

2. Application Development (10 Marks)

- **User Interface Design:**
 - Initial design of user-friendly UI with planned features.
 - Smooth back and forth flow between the pages of application.
- **Frontend-Backend Integration:**
 - Fully functional frontend interacting with backend systems.
 - Partial Front-end UI Designs to handle AI Model Deployment.
- **Backend Development:**
 - Robust backend architecture for data storage, processing, and system operations.
- **Workflow Implementation:**
 - Complete flow for data input and output display.

Proposed Modules

- **User Authentication Module:** Handles login, registration, and access control.
- **Landing Page/Home Page Module:** Welcomes users with an overview of the application's features, branding, and navigation.
- **Dashboard Module:** Allows users to interact with the application, performing basic operations which trigger changes in UI and backend seamless.

3. AI Model Integration (5 Marks)

- **Model Development:**
 - Selection and integration of a pre-trained AI model (if applicable).
 - Conduct data cleaning and initial training on mock or real-time data, sharing results to validate model viability.
 - Demonstration of the model's basic functionality.

4. Integration with Information System (5 Marks)

- **Initial Data Flow Implementation:** Ensure a working data flow between the mobile app and backend, but with a primary focus on model performance rather than complex system integration.

- **Mock Information System Setup:**
 - A simple backend that stores model predictions, focusing more on testing and logging model outputs.
- **Workflow Design for Integration:**
 - High-level design, indicating how model predictions will flow into the app, and how the backend will store these results.

5. Documentation and Technical Design (5 Marks)

- **Updated System Architecture Diagram:**
 - Diagram highlighting the AI model at the core and how it integrates with the application.
- **Workflow and Sequence Diagrams:**
 - Visualize data flow mainly focusing on model input and output.
- **Updated Requirements Document:**
 - Focus on requirements for training the model, its integration into the app, and backend handling of the model results.

6. Prototype Demonstration (5 Marks)

- **Partially functional application:**
 - User interface supporting interactions.
 - Backend implementation and interaction with front-end on real time data
 - Dataset Information, Data Cleaning and Partial Training of model with training techniques information.

Project Management (5 Marks)

Timeline Adherence:

- Milestones achieved as planned for the first 30% phase.
- Evidence of timely completion of key deliverables.

Collaboration and Communication:

- Regular meetings and feedback with supervisors documented.
- Example: Meeting notes, supervisor approvals, or comments on progress.

Use of Project Tracking Tools:

- Use of tools like Jira, Trello, or GitHub for issue tracking and progress updates.

Hybrid-Centric Approach

Focus: Equal emphasis on both the AI model and the application.

1. Proposal Refinement and Validation (5 Marks)

- Balanced objectives covering AI model development and application functionality.
- Feasibility studies for both AI and application workflows.
- Updated proposal incorporating feedback from the defense.

2. AI Model Integration and Development (5 Marks)

- **Pre-trained Model Usage (if applicable):**
 - Selection and rationale for the chosen model.
 - Fine-tuning or adapting the model for specific tasks. (if applicable)
 - Testing the pre-trained model on various mock scenarios to validate response quality and context understanding.
- **Core AI Functionality:**
 - Conduct **initial training** and testing on mock or real-time data, sharing results to validate model viability, even if not fully implemented in the application yet.
 - Example: Predictions or recommendations on test data.
- **Data Preprocessing Pipeline:**
 - Early-stage data cleaning and preparation for input data.

3. Application Development (10 Marks)

- **User Interface and Interaction:**
 - Interface with proper navigation between app pages and sections, ensuring seamless updates with real-time data changes triggered by user operations
 - Partial front-end setup for handling AI Integration and Deployment.
- **Backend and Data Flow:**
 - Data flow from app → backend system.
 - Establish functional connections between the mobile app frontend and backend services, utilizing either mock or real API calls for data exchange.

4. Integration with Information System (5 Marks)

- **Initial Data Flow Implementation:**
 - Implement data flow from the app to the backend.
- **Mock Information System Setup:**
 - Set up a mock or partial real - backend to simulate real-world interactions between the mobile app.

- **Workflow Design for Integration:**

- Design workflows that demonstrate how the app, AI module, and backend will interact and function together.

5. Documentation and Technical Design (5 Marks)

- Architecture diagram showing equal emphasis on AI and application components.
- Workflow diagrams for both AI processes and application interactions.
- Refined requirements document focusing on both aspects.

6. Prototype Demonstration (5 Marks)

- **Partially functional prototype demonstrating:**
 - AI model training and testing with performance benchmarks.
 - User interaction with the interface.
- **End-to-end workflow for data input and storage.**

7. Project Management (5 Marks)

Timeline Adherence:

- Milestones achieved as planned for the first 30% phase.
- Evidence of timely completion of key deliverables.

Collaboration and Communication:

- Regular meetings and feedback with supervisors documented.
- Example: Meeting notes, supervisor approvals, or comments on progress.

Use of Project Tracking Tools:

- Use of tools like Jira, Trello, or GitHub for issue tracking and progress updates.